

Group Number: 9

Group members: Kenny Yang, Ivan Shao, Minh Ly

Lab Preparation

Introduction

All labs in this course will run in the Remote Lab Emulation System (RLES). To prepare for Lab 1, this assignment asks your group to plan the required infrastructure. Map out your devices, IP addressing scheme, and domain information. Use the sample topology in Figure 1 as a guide.

Complete this document and submit a printed copy to the instructor at the end of the first lab session.

Domains

You will need two domains: one for Windows and one for Linux. Create descriptive domain names for each environment and record them below.

- **Linux Domain** linux.lab
- **Windows Domain** windows.lab

Servers

You will need three servers for the first lab. Below is a summary of each server's role. For the Linux and Windows domain controllers, you will need to assign a static IP address and hostnames.

Table 1 – Server Roles Summary

Windows Server 2025	Red Hat Enterprise Linux 9	Wiki Server
Active Directory Server	Identity Management Server	Rocky/Windows
Primary DNS Server	Primary DNS Server	
DHCP Server		

Record the server hostname information below.

Server Hostnames

- Windows Server 2025 win-dc01

- Red Hat Enterprise Linux (RHEL) rhel-idm01
- Wiki Server rocky-wiki01

Server Static IP Assignment

Using standard IPv4 conventions, choose a subnet for each network. Reserve the first ten low addresses in each subnet for servers. Also reserve three high-end addresses for the gateway—you'll use one now and keep the others in reserve for future needs.

Server IPv4 Static Address Assignment

- Windows Server 2025 192.168.10.2
- Red Hat Enterprise Linux Server 192.168.20.5
- Wiki Server 192.168.20.6

Gateway (pfSense)

The pfSense router will serve as the default gateway for both internal networks and the WAN edge to the RLES cloud. Assign static IP addresses to the two internal interfaces (one per network), and configure the WAN interface to obtain its address via DHCP. You may configure pfSense to act as the DHCP relay. Record the requested information below.

- Linux Interface IP address 192.168.20.254
- Windows Interface IP address 192.168.10.254
- pfSense (FQDN)

Clients

The first lab requires two clients, one for each domain. Record the clients' hostname information below.

- Linux Client _____
- Windows Client _____

Please Note: This is a sample topology. You may deploy your Wiki in the Windows domain, the decision is up to your group.

Figure 1 – Sample Topology

